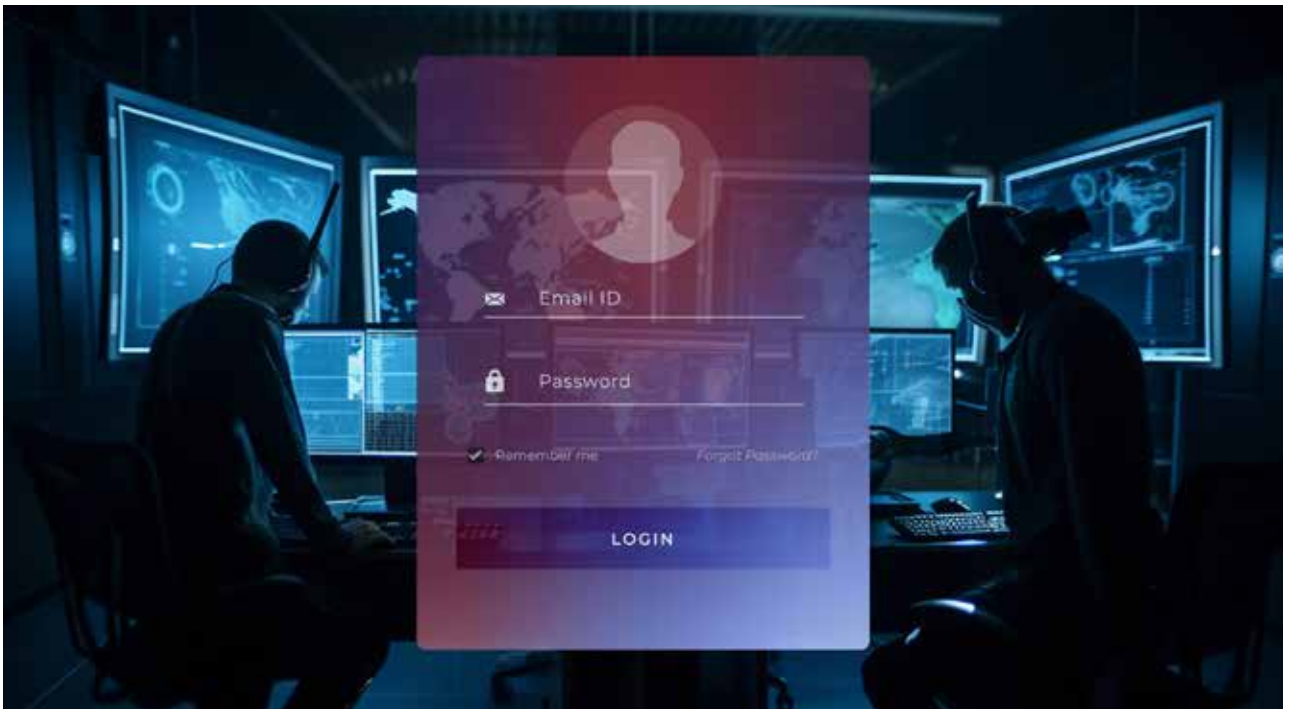


SageMath: A Quick Introduction to Cybersecurity

In the previous articles in this SageMath series, we delved into graph theory and explored its applications using SageMath. In this seventh article in the series, it is time to shift our focus to another crucial subfield of computer science: cybersecurity and cryptography.



Before diving deeper into the technical discussion, it is important to clarify a key point. While cybersecurity is a popular buzzword in the IT industry, we need to compare three widely used terms: digital security, cybersecurity, and cryptography. How are they related? There are varying viewpoints on the relationship between these fields. Figure 1 represents this relationship as a Venn diagram, illustrating how these concepts are related.

Digital security is the broadest of the three and encompasses all measures

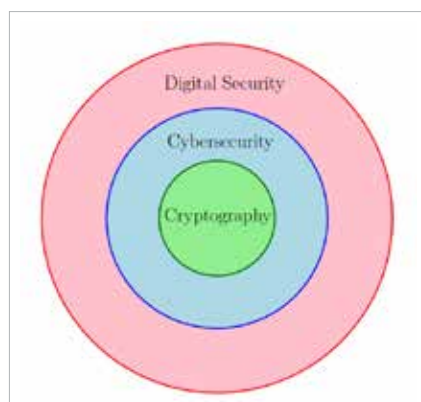


Figure 1: The relationship between digital security, cybersecurity, and cryptography

aimed at protecting digital assets and information. This includes strategies, technologies, and practices designed to safeguard digital data, devices, and systems from unauthorised access, damage, or loss, whether online or offline.

Cybersecurity is a more focused subset of digital security, dedicated specifically to defending digital systems and data from threats that arise in cyberspace. It involves protecting computer systems, networks, and online resources from cyberattacks, such as hacking, malware, phishing, and other forms of cybercrime.